PRECALCULUS · UNIT 7: VECTORS

Flight Path Navigator

Use vectors to fly a small plane through wind, find ground speed, and plan a delivery route.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You are the flight planner for Coastal Air Delivery. Small planes fly between islands, and the wind pushes them off course. Pilots need to know which way to point the plane and how long each leg will take.

You will use vectors to plan safe, efficient flights.



Plane
Airspeed 150 mph

Wind
30 mph from the west
(blowing east)

Learning Goals

- I can write vectors in component form and find magnitude and direction.
- I can add vectors to find a resultant.
- I can interpret ground speed and course.

Project Timeline

When	What you will do
Day 1	Vectors on the Map
Day 2	Wind
Day 3	Plan a Flight
Final	Flight Plan + reflection

Part 1: Vectors on the Map

Day 1 · Use the Flight Data Cards.

Flight Data Cards (x = east, y = north, miles)

Mainland Airport	Gull Island	Pine Island
(0, 0)	(120, 90)	(200, −50)

Wind A	Wind B	Cargo Ramp
⟨30, 0⟩ mph	25 mph toward bearing 135° (SE)	Pull 80 lb at 25° for 40 ft

1. Write the vector from the Mainland Airport to Gull Island in component form.

   Starter: $\langle x_2 - x_1,\ y_2 - y_1 \rangle$

2. Find its magnitude (the distance in miles).

3. Find its direction angle from east.

4. Write the vector from Gull Island to Pine Island.

Part 2: Wind

Day 2 · The plane flies due north at 150 mph. The wind is $\langle 30, 0 \rangle$.

5. Write the plane's air velocity as a vector.

6. Add the wind vector to find the ground velocity.

7. Find the ground speed.

8. By how many degrees does the wind push the plane off course?

Part 3: Plan a Flight

Day 3 · Use scalar multiplication.

9. If the ground velocity is $\langle 30, 150 \rangle$ mph, where is the plane after 2 hours?

10. Find a unit vector in the direction of Gull Island.

11. Draw a vector diagram of one flight and label the air velocity, wind, and ground velocity.

Final Product: Flight Plan

Your final product must include:

- Map vectors with magnitudes and directions
- Wind resultant and ground speed
- Position after a flight
- Labeled vector diagram
- Flight plan

Reflection

R1. What does the resultant vector represent for the plane?

R2. Why can the ground speed be faster than the airspeed?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Vectors, resultants, headings, and dot products are computed correctly and produce a workable flight plan. Goes beyond the minimum with a creative, realistic choice.	Vectors, resultants, headings, and dot products are computed correctly and produce a workable flight plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Flight plan includes diagrams, components, headings, speeds, and times. Polished, organized, and visually engaging.	Flight plan includes diagrams, components, headings, speeds, and times. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.